

Oscar Jaroker

Chatham, NJ | oscarjaroker@gmail.com | oscar.jaroker.com | linkedin.com/in/oscar-jaroker/

Education

Colgate University, BA in Applied Mathematics and Physics Aug 2023–May 2027

- GPA: 3.96 / 4.00 Overall; 4.00 / 4.00 in Majors
- **Coursework:** Data Analysis, Statistical Learning, Combinatorics, Real Analysis, Lagrangian Mechanics, Nonlinear Dynamics & Chaos, Quantum Mechanics, Electronics, Object-Oriented Programming, Economics
- **Awards and Honors:** Edwin Foster Kingsbury Prize for “most outstanding” physics student (May 2024)
Dean’s Award with Distinction (All Semesters)

Experience

Undergraduate Research and Teaching Roles, Colgate University – Hamilton, NY

- **Undergraduate Research Assistant**, Geography Laboratory Sept 2025–Present
Built and programmed Arduino-based data collection devices for actuator movement; integrated CH₄ and CO₂ sensors with I²C communication protocols
- **Laboratory Teaching Assistant**, Digital and Analog Electronics Jan 2026–Present
Supported lab sessions by assisting students with experiments, troubleshooting circuits, and grading assignments
- **Course Teaching Assistant**, Introductory Quantum Mechanics and Special Relativity Jan 2026–Present
Led homework discussion sections and held office hours for undergraduate physics course

Teaching Assistant, Art of Problem Solving – Millburn, NJ July 2025–Aug 2025

- Prepared students for the Science Olympiad through hands-on demonstrations and experiments in optics, circuits, mechanics, robotics, electricity and magnetism
- Guided 18 students through complex concepts and problem-solving, fostering collaboration and critical thinking

Corporate Treasury Intern, Pfizer – New York, NY June 2024–Aug 2024

- Conducted financial research of pharmaceutical industry by coding a drug-industry benchmarking program, accessing Bloomberg data through Python API, to compare key company metrics across peer groups
- Developed pricing simulator for interest-rate swaps using Bloomberg API and R to understand how interest rate movements might affect Pfizer’s portfolio
- Communicated results through multiple PowerPoint presentations to Capital Markets, Capital Strategy, Risk Management groups to show how company metrics and potential movements could impact future strategy

Instructor, International Ivy Summer Enrichment Programs – Summit, NJ June 2023–Aug 2023

- Taught Arduino and Makey Makey courses by building off of the existing curriculum to provide half-day instruction to students ages 8-14, developing their collaboration, design, and programming skills
- Awarded bonus as a top instructor based on positive feedback from students and parents

Academic Projects & Activities

“Optimal Harvesting in Stochastic Predator-Prey Environment” June 2025–Present

- Applied stochastic control to find optimal harvesting in 2-species fishery following highly nonlinear dynamics
- Used MATLAB to implement numerical methods such as FB Sweep, Euler-Maruyama, Backward Euler

President of Colgate Engineering and Design Club February 2025–Present

- Founded organization to build interesting engineering innovation by recruiting interested members, working with Colgate Office of Student Involvement and Physics Department, and completing BAC & leadership training

RIT Physics Olympiad 2024 & 2025

- Won second place out of 23 teams in 2025 and received a cash prize

Skills

Programming: Python (Pandas, Matplotlib), Java, MATLAB, R (tidyr, ggplot2), SQL, C++ , Mathematica

Language: Spanish (Limited Proficiency)